



Jordan matrix algebras

Lemma 1: Let J be finite-dimensional, simple, reduced Jordan algebra and $1 = e_1 + \dots + e_n$ be decomposition of unity element as a sum of absolutely primitive pairwise orthogonal idempotents.

Then, $J_{ij} \neq 0$ for all distinct $i, j \in \{1, \dots, n\}$.

Proof: First observe that if $J_{ij} \neq 0$ then $J_{ij}^2 \neq 0$.

Indeed, if $J_{ij}^2 = 0$ then $J_{ij} = Nil(Fe_i + J_{ij} + Fe_j) = 0$
 $= J_1(e_i + e_j)$

From the other side $Fe_i + J_{ij} + Fe_j$ is Jordan algebra of Clifford type from Proposition about Jordan algebras of capacity two.

Then if $J_{ij} \neq 0$, then there exist $a_{ij} \in J_{ij}$ with

$$a_{ij}^2 = \alpha(e_i + e_j), \quad \alpha \neq 0. \quad (*)$$

Take i, j, k distinct, for any $a_{ij} \in J_{ij}$ and $b_{jk} \in J_{jk}$

$$a_{ij}^2 b_{jk} = -2(a_{ij}^2, e_k, b_{jk}) = 4(a_{ij} b_{jk}, e_k, a_{ij}) = 2(a_{ij} b_{jk}) a_{ij}$$

$\{ (a^2, b, c) + 2(ac, b, a) = 0 \}$

$$\Rightarrow a_{ij}^2 b_{jk} = 2(a_{ij} b_{jk}) a_{ij} \quad (**)$$

Next we show that if $J_{ij} \neq 0$ and $J_{jn} \neq 0$ then $J_{ik} \neq 0$.

By (*) there exist $a_{ij} \in J_{ij}$ and $b_{jk} \in J_{jk}$ such that

$$a_{ij}^2 = \alpha(e_i + e_j) \quad \text{and} \quad b_{jk}^2 = \beta(e_j + e_k), \quad \text{then}$$

$$2(a_{ij} b_{jk})^2 = -a_{ij}^2 b_{jk}^2 + (a_{ij}^2 b_{jk}) b_{jk} + 2((a_{ij} b_{jk}) b_{jk}) a_{ij} =$$

$\{ (a^2, b, c) + 2(ac, b, a) = 0 \}$

$$= -\alpha\beta(e_i + e_j)(e_j + e_k) + \alpha((e_i + e_j) b_{jk}) b_{jk} + (a_{ij} b_{jk}^2) a_{ij} =$$

$$= -\alpha\beta e_i + \frac{1}{2} \alpha b_{jk}^2 + \frac{1}{2} \beta a_{ij}^2 = \frac{1}{2} \alpha\beta (e_i + e_k) \neq 0.$$

Then $J_{ik} \neq 0$.

Define relation $\approx$ in $\{1, \dots, n\}$ $i \approx j$ iff $J_{ij} \neq 0$

Then it is equivalence relation (only non-trivial transitivity is equivalence relation)



Let $S_1, \dots, S_k$ be equivalence classes of $\{1, \dots, n\}$ with respect to $\approx$. Let $E_i = \sum_{e \in S_i} e$ for $i=1, \dots, k$, then

$E_1, \dots, E_k$ are pairwise orthogonal idempotents, with $1 = E_1 + \dots + E_k$ and $J_l(E_l) = \sum_{i \in S_l} F_{ei} + \sum_{\substack{ij \in S_l \\ i < j}} J_{ij}$ for all $l=1, \dots, k$.

Then $J = J_1(E_1) \oplus \dots \oplus J_1(E_k)$ and each $J_1(E_l) \triangleleft J$. Since J is simple we prove that $c=0 \Rightarrow$ all $J_{ij} \neq 0$.

Def: For J defined in previous Lemma, tell that e_i and e_j are connected if $\exists a_{ij} \in J_{ij}$ invertible in $J_1(e_i + e_j)$. e_i and e_j are strictly connected if $\exists a_{ij} \in J$ with $a_{ij}^2 = e_i + e_j$.

Corollary 1: For J satisfying conditions of Lemma 1 and $F = \overline{F}$: for any $i \neq j \in \{1, \dots, n\}$ the idempotents e_i and e_j are strictly connected.

Proof: We have seen in the proof of Lemma 1 $\exists a_{ij} \in J_{ij}$ such that $a_{ij}^2 = \alpha(e_i + e_j)$, $\alpha \in F$, $\alpha \neq 0$. Since $F = \overline{F}$ $\exists \beta \in F$ with $\beta^2 = \alpha$: $(\beta^{-1} a_{ij})^2 = \beta^{-2} \alpha (e_i + e_j) = e_i + e_j$.

Lemma 2: Let J be Jordan algebra satisfying Lemma 1, then there exists $e_{ij} \in J_{ij}$, $i \neq j$ such that
(i) $e_{ij}^2 = e_i + e_j$ and
(ii) $2e_{ij}e_{jk} = e_{ik}$ for each $k \neq i, j$.

Proof:

By Corollary 1 e_1 and e_i $i \neq 1$ are strictly connected

$\Rightarrow \exists e_{1i} \in J_{1i}$ $e_{1i}^2 = e_1 + e_i$. Define

$e_{ij} = 2e_{1i}e_{1j}$ for all $i, j > 1$, $i \neq j$.

Then $e_{ij}^2 = 4(e_{1i}e_{1j})^2 = -2e_{1i}^2e_{1j}^2 + 2(e_{1i}^2e_{1j})e_{1j} + 4(e_{1i}e_{1j})e_{1j}e_{1i}$

$$\begin{aligned} 1LJI &= \{ (a^2, b, c) + 2(ac, b, a) = 0 \} \quad (**) \\ &= -2e_{1i}^2e_{1j}^2 + 2(e_{1i}^2e_{1j})e_{1j} + 2(e_{1j}^2e_{1i})e_{1i} \end{aligned}$$

$$= -2(e_1 + e_i)(e_1 + e_j) + 2((e_1 + e_i)e_{1j})e_{1j} + 2((e_1 + e_j)e_{1i})e_{1i}$$

$$= -2e_1 + e_{1j}^2 + e_{1i}^2 = -2e_1 + e_1 + e_j + e_1 + e_i = e_i + e_j$$

For distinct i, j, k :

if $j=1$ $2e_{ij}e_{jk} = e_{ik}$ by definition

if $i=1$ (or $k=1$) we have $2e_{ij}e_{jk} = 2e_{1j}(2e_{1j}e_{1k}) =$

$$= 4(e_{1k}e_{1j})e_{1j} = 2e_{1k}e_{1j}^2 = 2e_{1k}(e_1 + e_j) = e_{1k};$$

if i, j, k distinct from 1!



$$\begin{aligned} 2e_{ij}e_{jk} &= \delta(e_{ii}e_{ij})(e_{jj}e_{jk}) = -4e_{ij}^2(e_{ii}e_{jk}) + 4(e_{ij}^2e_{ii})e_{jk} \\ &+ \delta((e_{ij}e_{jk})e_{ii})e_{ij} = -2(e_i+e_j)e_{ik} + 4((e_i+e_j)e_{ii})e_{ik} + \\ &+ 4(e_{jk}e_{ii})e_{ij} = 2e_{ii}e_{ik} = e_{ik}. \end{aligned} \quad \square$$

let J be finite-dimensional, simple Jordan algebra of capacity > 2 and $F = \bar{F}$. let $e_1, \dots, e_n$ be primitive idempotents, pairwise orthogonal, $1 = e_1 + \dots + e_n$.
By Proposition about simple Jordan of capacity 2 for $i+j$
 $J_1(e_i+e_j) = Fe_i + J_{ij} + Fe_j$ are subalgebras of Clifford type

let $f_{ij}: J_{ij} \rightarrow F$ such that $a_{ij}^2 = f_{ij}(a_{ij})(e_i+e_j)$,
for any $a_{ij} \in J_{ij}$.

Then f is quadratic, since $f_{ij}(\alpha a_{ij}) = \alpha^2 a_{ij}^2 = \alpha^2 f_{ij}(a_{ij})$
 $\forall \alpha \in F, \forall a_{ij} \in J_{ij}$

let $\bar{f}_{ij}$ be a bilinear form associated to quadratic f_{ij}
 $\bar{f}_{ij}(a_{ij}, b_{ij}) = \frac{1}{2}(f_{ij}(a_{ij}+b_{ij}) - f_{ij}(a_{ij}) - f_{ij}(b_{ij}))$

then

$$\bar{f}_{ij}(a_{ij}, b_{ij})(e_i+e_j) = \frac{1}{2}((a_{ij}+b_{ij})^2 - a_{ij}^2 - b_{ij}^2) = a_{ij}b_{ij}.$$

Then $\bar{f}_{ij}$ is non-degenerate: since $\forall a_{ij} \in J_{ij}$ such that $\bar{f}_{ij}(a_{ij}, \bar{a}_{ij}) = 0$
we get $a_{ij}J_{ij} = 0$ and next $Fe_{ij} \triangleleft J_1(e_i+e_j)$ with $(Fe_{ij})^2 = 0$
then $a_{ij} = 0$.

Lemma 3: let J be fin-dim simple Jordan algebra, $F = \bar{F}$
of capacity $n > 2$ ($1 = e_1 + \dots + e_n$). Let i, j, k
be all distinct. let f_{ij}, f_{ik}, f_{jk} be
quadratic forms for $J_1(e_i+e_j), J_1(e_i+e_k)$ and $J_1(e_j+e_k)$

Then $f_{ik}(2a_{ij}b_{jk}) = f_{ij}(a_{ij})f_{jk}(b_{jk}), \forall a_{ij} \in J_{ij}, b_{jk} \in J_{jk}$

$$\begin{aligned} \text{Proof: By 1LJI } 2(a_{ij}b_{jk})^2 &= \\ &= -a_{ij}^2 b_{jk}^2 + (a_{ij}^2 b_{jk})b_{jk} + 2((a_{ij}b_{jk})b_{jk})a_{ij} = \\ &= -a_{ij}^2 b_{jk}^2 + (a_{ij}^2 b_{jk})b_{jk} + (b_{jk}^2 a_{ij})a_{ij} = \text{definition of } f \\ &= -f_{ij}(a_{ij})(e_i+e_j)f_{jk}(b_{jk})(e_j+e_k) + f_{ij}(a_{ij})((e_i+e_j)b_{jk})b_{jk} + \\ &+ f_{jk}(b_{jk})((e_k+e_j)a_{ij})a_{ij} = -f_{ij}(a_{ij})f_{jk}(b_{jk})e_i + \\ &+ \frac{1}{2}f_{ij}(a_{ij})f_{jk}(b_{jk})(e_j+e_k) + \frac{1}{2}f_{jk}(b_{jk})f_{ij}(a_{ij})(e_i+e_j) = \end{aligned}$$



$$= f_{ij}(a_{ij}) f_{jk}(b_{jk}) (-e_j + \frac{1}{2}(e_j + e_k) + \frac{1}{2}(e_i + e_j)) = \frac{1}{2} f_{ij}(a_{ij}) f_{jk}(b_{jk}) (e_i + e_k)$$

From the other side, since $a_{ij} b_{jk} \in J_{ik}$, we get

$$2(a_{ij} b_{jk})^2 = 2 f_{ik}(a_{ij} b_{jk}) (e_i + e_k) = \frac{1}{2} f_{ik}(2 a_{ij} b_{jk}) (e_i + e_k)$$

since f_{ik} is quadratic. $\square$

Lemma 4: Under assumption of Lemma 3 all subspaces J_{ij} with $i \neq j$ are isomorphic as vector spaces.

Proof: linear mapping $\varphi_{ijk}: J_{ij} \rightarrow J_{ik}$ defined by $\varphi_{ijk}(a_{ij}) = a_{ij} e_{jk}$ is injective.

Indeed, suppose $a_{ij} e_{jk} = 0$ for some $a_{ij} \in J_{ij}$

$$0 = 2(a_{ij} e_{jk}) e_{jk} = \underset{(**)}{e_{jk}^2} a_{ij} = (e_j + e_k) a_{ij} = \frac{1}{2} a_{ij}$$

Then φ_{ijk} is injective for all i, j, k . Analogously $\varphi_{kji}: J_{kj} = J_{jk} \rightarrow J_{ji} = J_{ij}$ is injective then we get

the following inequality $\dim J_{ij} \leq \dim J_{jk} \leq \dim J_{ji} \Rightarrow$

all J_{ij} have the same dimension $\Rightarrow$ isomorphic.

Def: We call an algebra A a composition algebra if A has unity and a quadratic ~~form~~ non-degenerate form n in A such that $n(xy) = n(x)n(y)$, $x, y \in A$.

Remark: Any composition algebra A is quadratic, this is $\forall a \in A$

$$a^2 = t(a)a - n(a)1, \text{ with } t(a), n(a) \in F$$

and has involution $a \rightarrow \bar{a}$ such that

$$a + \bar{a} = t(a)1 \text{ and } a\bar{a} = \bar{a}a = n(a)1 \quad \forall a \in A.$$

Proposition 1: Let J be Jordan algebra satisfying conditions of Lemma 3 with e_{ij} defined as in Lemma 2.

Consider $D = J_{12}$ with the following multiplication

$$a * b = f(a e_{23})(b e_{13}) \text{ for } a, b \in D$$

and denote f_{12} by n . Then

$$n(a * b) = n(a)n(b) \text{ and } a * e_2 = e_2 * a = a$$

$\forall a, b \in D$. Then D is a composition algebra with quadratic form n and unity e_2 .

Proof: We already saw that J_{12} is non-degenerated



$$n(a \times b) = f_{12}(f(ae_{23})(be_{13})) = f_{13}(2ae_{23}) f_{23}(2be_{13})$$

$$= f_{12}(a) \underbrace{f_{23}(e_{23})}_{=1} f_{12}(b) \underbrace{f_{13}(e_{13})}_{=1} = n(a)n(b) \quad (f_{ij}(e_j) = 1 \text{ for } i \neq j)$$

Moreover

$$a \times e_{12} = f(ae_{23})(e_{12}e_{13}) = 4(ae_{23})e_{23} = (**) \\ = 2ae_{23}^2 = 2a(e_2 + e_3) = a \quad \text{and}$$

$$e_{12} \times a = f(e_{12}e_{23})(ae_{13}) = 4e_{13}(ae_{13}) = (**) = 2ae_{13}^2 = 2a(e_1 + e_3) = a$$

$\forall a \in D \Rightarrow D$ is a composition algebra. $\square$

We have shown that $J_{12} \cong D$ is a composition algebra with unit element e_{12} and quadratic form $n = f_{12}$.

We will show that $t(a) = tr(a)(e_1 + e_2) = 2ae_{12} \quad \forall a \in D$.

$$\text{Let } a \in D \quad a \times a = f(ae_{23})(ae_{13}) = (ac, b, d) + (ad, b, c) + (cd, b, a) = 0$$

$$+ 4((ae_{13})a)e_{23} + 4a(ae_{13}e_{23}) = (**) \\ = -2n(a)(e_1 + e_2)e_{12} + 2(a^2e_{23})e_{13} + 2(a^2e_{13})e_{23} + 2a(ae_{12}) = (**) \\ = -2n(a)e_{12} + 2(n(a)(e_1 + e_2)e_{23})e_{13} + 2(n(a)(e_1 + e_2)e_{13})e_{23} + 2(ae_{12})a$$

$$= n(a)(-2e_{12} + \frac{1}{2}e_{12} + \frac{1}{2}e_{12}) + 2(ae_{12})a = 2(ae_{12})a - n(a)e_{12} =$$

$$= 2(ae_{12})a - n(a)1_D$$

As D is quadratic we have $2ae_{12} \in F1$ and then it follows that $t(a)(e_1 + e_2) = 2ae_{12}$.

Additionally we have $\bar{a} = t(a)1_D - a = 2(ae_{12})e_{12} - a \quad \forall a \in D$

Let A be an algebra with involution $a \mapsto \bar{a}$. We may extend this involution to a matrix algebra

$$M_n(A) \quad \overline{\{a_{ij}\}} = \{\bar{a}_{ji}\}, \text{ for any } \{a_{ij}\} \in M_n(A)$$

We call a matrix $x \in M_n(A)$ Hermitian if $\bar{x} = x$.

The set of Hermitian matrices is denoted by $H_n(A)$.

In fact $H_n(A)$ is a subalgebra of $M_n(A)^+$ since for any $x, y \in H_n(A)$ $x \cdot y = \frac{1}{2}(xy + yx) \in H_n(A)$.

For $a \in A$ and $i, j = 1, \dots, n$ define $a[ij] = ae_{ij} + \bar{a}e_{ji}$

(here e_{ij} denotes elementary matrix with only 1 on entry (i, j) while other entries zero)



We have $a[ij] \in U_n(A)$ and $a[ij] = \bar{a}[ji]$.

$$A \hookrightarrow M_n(A) \quad a \mapsto \begin{bmatrix} a & & \\ & \ddots & \\ & & a \end{bmatrix}$$

If $x \in U_n(A)$ then x has a form

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \bar{a}_{12} & & & \\ \vdots & & \ddots & \\ \bar{a}_{1n} & & & a_{nn} \end{bmatrix}$$

with $\bar{a}_i = a_i \quad i=1, \dots, n$

For each $i=1, \dots, n$ set $e_i = e_{ii}$, then

$e_1, \dots, e_n$ are pairwise orthogonal idempotents $1 = e_1 + \dots + e_n$

Then the full Peirce decomposition

$$U_n(A) = \sum_{i=1}^n U_n(A)_{ii} + \sum_{1 \leq i < j \leq n} U_n(A)_{ij}$$

For distinct i, j we have $U_n(A)_{ii} = \{a[ii] : a \in A\}$

$$U_n(A)_{ij} = \{a[ij] : a \in A\}$$

Indeed $a[ii] = (a + \bar{a})e_{ii}$ and

$$\begin{aligned} a[ij] \cdot e_i &= \frac{1}{2} (a e_{ij} + \bar{a} e_{ji}) e_i + e_i (a e_{ij} + \bar{a} e_{ji}) = \\ &= \frac{1}{2} (\bar{a} e_{ji} + a e_{ij}) = \frac{1}{2} a[ij] \end{aligned}$$

Lemma 5: let $i, j, k \in \{1, \dots, n\}$ distinct. Then

$$(i) \quad 2a[ij] \cdot b[jk] = (ab)[ik];$$

$$(ii) \quad 2a[ii] \cdot b[ik] = (a + \bar{a})b[ik];$$

$$(iii) \quad 2a[ij] \cdot b[jj] = a(b + \bar{b})[ij];$$

$$(iv) \quad 2a[ij] \cdot b[ij] = (a\bar{b})[ii] + (\bar{a}b)[jj];$$

$$(v) \quad 2a[rs] \cdot b[kl] = 0 \text{ if } \{r, s\} \cap \{k, l\} = \emptyset$$

Proof: the proof is straightforward. For example

$$\begin{aligned} (i) \quad (a e_{ij} + \bar{a} e_{ji})(b e_{jk} + \bar{b} e_{kj}) + (b e_{jk} + \bar{b} e_{kj})(a e_{ij} + \bar{a} e_{ji}) &= \\ = (ab) e_{ik} + (\bar{b}\bar{a}) e_{ki} = (ab) e_{ik} + (\overline{ab}) e_{ki} = (ab)[ik] \end{aligned}$$

$$\begin{aligned} (iv) \quad (a e_{ij} + \bar{a} e_{ji})(b e_{ij} + \bar{b} e_{ji}) + (b e_{ij} + \bar{b} e_{ji})(a e_{ij} + \bar{a} e_{ji}) &= \\ = a\bar{b} e_{ii} + \bar{a} b e_{jj} + \bar{b}\bar{a} e_{ii} + b a e_{jj} = (a\bar{b})[ii] + (\bar{a}b)[jj] \end{aligned}$$

Def: An algebra A is called alternative if

$$(a, a, b) = 0 \text{ and } (a, b, b) = 0 \text{ for all } a, b \in A.$$



Theorem 1: Let $n \geq 3$. If $U_n(A)$ is a Jordan algebra, then

A is an alternative algebra (associative, if $n \geq 4$)
and for any $a \in A$, $a + \bar{a} \in N(A)$, the center of A .

Proof: For $i, j \in \{1, \dots, n\}$ we have shown that
 $(U_n(A))_{ij} = \{a[ij] : a \in A\}$ with idempot. $e_i = \frac{1}{2}[ii]$.

Then we also have $(\star) (J_{ij}, J_{jk}, J_{kl}) = 0$ for $\{k, l\} \cap \{i, j\} = \emptyset$,

$$\text{since } (a_{ij}, b_{jk}, c_{kl}) = 2(a_{ij}e_i, b_{jk}, c_{kl}) = -2(\underbrace{a_{ij}c_{kl}}_{=0}, b_{jk}, e_i) - \\ - 2(\underbrace{e_i c_{kl}}_{=0}, b_{jk}, a_{ij}) = 0$$

Thus for $a, b, c \in A$ and $\{k, l\} \cap \{i, j\} = \emptyset \Rightarrow (a[ij], b[jk], c[kl]) = 0$

In particular $4(a[ij] \cdot b[jk]) \cdot c[kl] = 2(ab)[ik] \cdot c[kl] = ((ab)c)[il]$

$$4a[ij] \cdot (b[jk] \cdot c[kl]) = (a(bc))[il]$$

$$\Rightarrow (ab)c = a(bc). \text{ If } n > 3 \Rightarrow A \text{ is associative.}$$

Using $(\star)$ for $l=k$ we obtain $(J_{ij}, J_{jk}, J_{kk}) = 0$.

Then for $a, b, c \in A$ and $k \notin \{i, j\}$ we have

$$(a[ij], b[jk], c[kk]) = 0, \text{ rewriting}$$

both sides we get

$$4(a[ij] \cdot b[jk]) \cdot c[kk] = 2(ab)[ik] \cdot c[kk] = ((ab)(c + \bar{c}))[ik]$$

$$4a[ij] \cdot (b[jk] \cdot c[kk]) = 2a[ij] \cdot b(c + \bar{c})[jk] = (a(b(c + \bar{c}))) [ik]$$

$$\Rightarrow (a, b, c + \bar{c}) = 0, \text{ analogously } (a + \bar{a}, b, c) = 0$$

Finally $(a_{ij}, b_{jj}, c_{jk}) = (a_{ij}, b_{jj}e_j, e_{jk}) = b_{jj} \cdot (a_{ij}, \underbrace{e_j}_{\text{as derivation}}, c_{jk}) +$

$$+ \underbrace{(a_{ij}, b_{jj}, c_{jk})}_{\in J_{ik}} e_j = 0$$

$\in J_{ik}$

$$\text{Thus } (a[ij], b[jj], c[jk]) = 0$$

$$\text{rewriting } (a, b + \bar{b}, c) = 0$$

Consequently $a + \bar{a} \in N(A)$.

I shall leave proof for $n=3$ and alternative A as exercise



Finally: Let J be finite-dimensional simple Jordan algebra with capacity > 2 and $F = \bar{F}$

$e_1, \dots, e_n$ primitive, pairwise orthogonal $1 = e_1 + \dots + e_n$

$e_{ij} \in J_{ij}$ such that $e_{ij}^2 = e_i + e_j$ and $2e_{ij}e_{jk} = e_{ik}$

Then by Prop. 1 $D = J_{12}$ is a composition algebra, i, j, k distinct

We will show that J is isomorphic to $H_n(D)$

For any element $a \in D = J_{12}$ we define $a_{ij} \in J_{ij}$ such

that $a[ij] \rightarrow a_{ij}$ determines an isomorphism

And for this we show that $a_{ij} \in J$ satisfy the same relations as $a[ij] \in H_n(D)$.

For each $a \in D$ and $i, j = 1, \dots, n$ $j \neq 1$ and $i \neq 2$ define

$$a_{ij} = 2a e_{2j}, \quad a_{j1} = \bar{a}_{1j}, \quad a_{i2} = 2a e_{i1}, \quad a_{2i} = \bar{a}_{i2}$$

$$a_{ii} = \text{tr}(a) e_i \quad \text{for } i = 1, \dots, n$$

and for $i, j > 2$ distinct

$$a_{ij} = 2a_{i2} e_{2j} = 4(a e_{i1}) e_{2j}.$$

Lemma 6: (i) for any $i \neq j$ $a_{ji} = \bar{a}_{ij}$

$$i, j, k \text{ distinct} \quad (i) \quad 2a_{ij}b_{jk} = (a \times b)_{ik}$$

$$(ii) \quad 2a_{ii}b_{ij} = \text{tr}(a)b_{ij};$$

$$\text{and } a, b \in D \quad (iii) \quad 2a_{ij}b_{jj} = \text{tr}(b)a_{ij};$$

$$J_{12}'' \quad (iv) \quad 2a_{ij}b_{ji} = \text{tr}(a \times b)(e_i + e_j) = (a \times b)_{ii} + (b \times a)_{jj}.$$

Theorem 2: Let J be a finite-dimensional simple reduced Jordan algebra of capacity $n \geq 3$, with n idempotents pairwise orthogonal, strictly connected.

Then $J \cong H_n(A, *)$, with A is a composition algebra, which is associative if $n > 3$.

Proof:

It is enough to show that $\varphi: a_{ij} \rightarrow a[ij], a \in A$ defines a bijection between vector spaces

J and $H_n(A, *)$. Moreover by Lemma 6

φ is an isomorphism of Jordan algebras.



Coordination theorem:

Let J be unital Jordan algebra such that
 $1 = e_1 + \dots + e_n$, with $e_1, \dots, e_n$ strictly connected,
pairwise orthogonal idempotents. Then

J is isomorphic to $M_n(A, *)$, with $(A, *)$ being
an alternative algebra (associative if $n \geq 4$) with
involution $*$ such that for every $a \in A$, $a * a^*$ is an
element of $N(A)$, the center of A .